

A categorical comparison of AU incompleteness, Beklemishev-Shamkanov, and Willard SJAS

R6 of the Refinement stage. Charter: [ADR-0004-lawvere-sjas-translation.md](#).

Status and scope. This is a comparison construction, not a claim that Beklemishev-Shamkanov’s calculus or Willard’s arithmetics are arithmetic universes. It identifies the categorical data used by the arithmetic-universe proof and then presents the other systems in the weakest common language that preserves their failure mechanisms. Claims about the three source corpora are page-image verified in [VERIFICATION.md](#).

0. Answer

The comparison needs **one more coordinate than the three Hilbert-Bernays-Löb (HBL) conditions**. In categorical notation, the relevant package is

1. a fixed point for the consistency-shaped endomorphism;
2. the three HBL structure maps for the provability operation $\Box$; and
3. a diagonal on boxed hypotheses, $\text{copy_A} : \Box A \rightarrow \Box A \text{ tensor } \Box A$, expressing $\Box$ -contraction.

The initial arithmetic universe has the whole package: finite products supply the diagonal, van Dijk-Oldenziel Propositions 6.5, 6.6 and 6.8 supply the HBL maps, and Lemma 6.12 supplies fixed points.

Beklemishev-Shamkanov’s example **S** does **not** fail one of the three HBL conditions. It satisfies their versions L1-L3 and has many fixed points. What it lacks is the boxed diagonal. Categorically, its Lindenbaum construction is symmetric monoidal rather than cartesian; $\Box A$ cannot in general be copied. Consequently the formalized-G2 arrow

$\Box(\Box \text{bottom} \multimap \text{bottom}) \rightarrow \Box \text{bottom}$

does not exist. But neither does a global proof of consistency $I \rightarrow (\Box \text{bottom} \multimap \text{bottom})$. Thus **S** blocks the G2 derivation without proving its own consistency.

Willard’s Type-A **IS(A)** lies on the other axis. Its ordinary classical logic has cartesian contraction, but its weak arithmetic cannot internalize the **uniform proof-code composition** required by HBL condition (2). In the syntactic proof doctrine there is no internal entailment

$\Box \Phi$ and $\Box(\Phi \rightarrow \Psi) \rightarrow \Box \Psi$

uniform in the two proof witnesses, although every required fixed-numeral instance can be verified. The same pointwise/uniform split lets **IS(A)** obtain the one substitution instance needed for its Group-3 fixed point without proving total substitution in general. Group-3 is a global self-consistency point, and Willard separately proves external consistency. That **positive pair**—internal self-consistency plus external consistency—is what the Beklemishev-Shamkanov toy system lacks.

In short:

System	Fixed point	HBL maps	Boxed diagonal	Internal self-consistency	Externally consistent
Joyal / van Dijk-Oldenziel AU	every $T : P1' \rightarrow P1'$	all three	yes, cartesian	impossible if U_0 is consistent	assumed in G2
Beklemishev-Shamkanov S	many modal fixed points	all three	no	no	not their advertised result
Willard Type-A IS(A)	selected Group-3 fixed point	HBL (2) fails uniformly	yes	yes	yes , under Proposition 1’s hypotheses (full)

This is the requested distinction: B-S establish **non-derivability of formalized G2**; Willard establishes a **consistent inhabitant of the self-consistency proposition**.

1. The common comparison object: a coded modal proof doctrine

Neither an arithmetic universe alone nor a bare modal algebra records all the distinctions at issue. We therefore use the following deliberately small interface. This is the Refinement's comparison construction, not terminology claimed from any of the three sources.

1.1 Structural category

Let K be a category, or a representable symmetric multicategory, of formula contexts and derivations. Its tensor **tensor** represents context combination, its unit **I** represents the empty context, and an internal implication $A \multimap B$, when present, satisfies

$$K(\text{Gamma tensor } A, B) \simeq K(\text{Gamma}, A \multimap B).$$

If **tensor** is a categorical product, every object has a diagonal

$$\text{delta}_A : A \rightarrow A \text{ tensor } A.$$

Proof-theoretically, precomposition with **delta_A** is contraction. In a merely monoidal category no such diagonal is automatic.

1.2 Provability data

For each coded sentence A , let **Prf_A** be its object or predicate of proof witnesses, and let

$$\Box A := \text{exists } p. \text{Prf}_A(p).$$

The three HBL conditions become the following structure.

Name	Categorical/proof-doctrine form	Classical condition
Nec	global proofs of A lift to global proofs of $\Box A$	if $\vdash A$, then $\vdash \Box A$
BoxMP	$m_{A,B} : \Box(A \multimap B) \text{ tensor } \Box A \rightarrow \Box B$	$\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$
Four	$\text{four}_A : \Box A \rightarrow \Box \Box A$	$\Box A \rightarrow \Box \Box A$

Across nonclassical logics, these three maps are not enough. The additional structural datum isolated by Beklemishev-Shamkanov is

$$\text{BoxContr}_A : \Box A \rightarrow \Box A \text{ tensor } \Box A.$$

In a cartesian doctrine this is just **delta_{ $\Box A$ }** and is invisible as a separate assumption. In an affine, linear, or other substructural doctrine it is real extra structure.

1.3 Fixed points and consistency points

A fixed point is an object or proposition G equipped with derivations both ways between G and $E(G)$. Three strengths must not be conflated:

- **universal**: every endomorphism in a stated class has a fixed point;
- **modalized**: every syntactically modalized formula has a selected fixed point;
- **selected**: the particular consistency-shaped endomorphism has a fixed point.

The G_2 argument needs only an appropriate selected fixed point. It does not follow that the object theory proves total diagonal substitution.

A *consistency point* is a global derivation

$$\text{con} : I \rightarrow \text{Con}, \text{ where } \text{Con} := \Box \text{bottom} \multimap \text{bottom}.$$

This is only the internal half of self-justification. The other half is the external, metatheoretic assertion that the theory is consistent. An inconsistent theory can of course inhabit every proposition, including **Con**.

1.4 The comparison theorem

Theorem 1.1 (categorical HBL-G2 package). Suppose the doctrine has an appropriate consistency-shaped fixed point, **Nec**, **BoxMP**, **Four**, boxed contraction, and enough implication rules to perform the usual propositional packaging. Then the formalized-G2 arrow $\Box\text{Con} \rightarrow \Box\text{bottom}$ is derivable; under external consistency there is no consistency point.

Proof. Let $G \leftrightarrow \text{not } \Box G$ be the fixed point. Apply **Nec** and **BoxMP** to the forward direction $G \rightarrow (\Box G \rightarrow \text{bottom})$. This gives

$$\Box G \rightarrow \Box(\Box G \rightarrow \text{bottom}).$$

Four gives $\Box G \rightarrow \Box\Box G$. To apply **BoxMP** once more, now with $A = \Box G$ and $B = \text{bottom}$, the same input $\Box G$ must feed both arrows. This is the one critical use of

$$\text{BoxContr}_G : \Box G \rightarrow \Box G \text{ tensor } \Box G.$$

The resulting composite is $\Box G \rightarrow \Box\text{bottom}$. Hence

$\text{Con} \rightarrow \text{not } \Box G$, and therefore $\text{Con} \rightarrow G$ by the reverse fixed-point arrow. If we apply **Nec** and **BoxMP** to $\text{Con} \rightarrow G$, we get $\Box\text{Con} \rightarrow \Box G$; composition therefore gives the advertised $\Box\text{Con} \rightarrow \Box\text{bottom}$. If **Con** had a global proof, **Nec** would supply $\Box\text{Con}$, formalized G2 would supply $\Box\text{bottom}$, and **Con** itself would yield **bottom**. This is the ordinary HBL proof with the normally implicit cartesian diagonal made explicit. Beklemishev-Shamkanov Proposition 3.8 identifies the same contraction use in their abstract proof. ■

The theorem is a **sufficient package**, not a biconditional. Failing any one component blocks this proof; it does not by itself manufacture a consistency point or prove external consistency.

2. Joyal / van Dijk-Oldenziel: the full cartesian package

2.1 Their actual AU construction

Let U_0 be the initial arithmetic universe and U'_0 the externalization of its internal initial arithmetic-universe object. Van Dijk-Oldenziel construct

- an interpretation functor $R : U_0 \rightarrow U'_0$;
- a global-sections functor $\Gamma : U'_0 \rightarrow U_0$; and
- the provability operation $\Box := \Gamma R$.

A proposition is a monomorphism $\phi \rightarrow 1$. Because an arithmetic universe is a list-arithmetic pretopos, its structural tensor is a finite product. Every proposition and every boxed proposition therefore has a diagonal.

The paper stresses that an arithmetic universe need not be cartesian closed. Section 6 consequently treats $\phi \vdash \psi$ as an inclusion of subobjects and interprets implications between such judgements in a classifying arithmetic universe. The comparison here follows that construction; it does not pretend that U_0 has an ordinary exponential implication.

2.2 HBL as AU arrows

The source's propositions instantiate the common interface exactly as follows.

Common datum	van Dijk-Oldenziel	Construction
Nec	Proposition 6.5	$U \models \phi$ implies $U \models \Box\phi$, from the unit-like map $\phi \rightarrow \Gamma\phi$ of Lemma 5.15
Four	Proposition 6.6	$U \models \Box\phi \vdash \Box\Box\phi$, by applying $\Box = \Gamma R$
BoxMP	Proposition 6.8	internal evaluation $\text{Hom}(\phi', \psi') \times \text{Hom}(1', \phi') \rightarrow \text{Hom}(1', \psi')$ in the classifying AU
ordinary MP	Propositions 6.7, 6.9, 6.10	composition and packaging of judgements
fixed point	Lemma 6.12	every $T : P1' \rightarrow P1'$ has a global fixed point

Common datum	van Dijk-Oldenziel	Construction
contraction	finite products in the AU	<code>delta_X : X -> X x X</code>

Thus Theorem 6.13 proves Löb. The direct G2 result is Theorem 5.20; equivalently, the false/consistency instance of the Löb package rules out an internal proof of consistency when U_0 is consistent.

The load-bearing map for comparison with the other systems is Proposition 6.8's evaluation. It is a single internal, uniform arrow. The existence of its value at every externally named pair would be weaker.

3. Beklemishev-Shamkanov: HBL without the boxed diagonal

3.1 The APS as a thin category

Beklemishev-Shamkanov Definition 2.1 starts with a preorder

$S = (L_S, \leq_S, \text{top}, \text{bottom})$.

View it as a thin category $L(S)$: its objects are sentences (or their deductive-equivalence classes), and there is a unique arrow $A \rightarrow B$ exactly when $A \leq_S B$. The operators $\Box$ (provability) and boxtimes (refutability) are object operations satisfying their APS conditions C1-C4. This is already an order-enriched categorical presentation of their abstract provability structure.

3.2 The consequence calculus as a monoidal category

Definitions 3.1-3.2 refine the preorder to multiset contexts. After quotienting by deductive equivalence, set

$K_S(A, B) = \{*\}$ if $A \vdash_S B$, and empty otherwise.

Multiset union is represented by their conjunction **tensor**; **top** is the unit; and **I3** gives the closure adjunction

$K_S(\text{Gamma tensor } A, B) \sim K_S(\text{Gamma}, A \rightarrow B)$.

Exchange is built in because contexts are multisets. Cut is categorical composition: for the concrete system S , Theorem 5 proves cut admissible, so this presentation really is compositional. What is absent is not composition but a cartesian diagonal.

Their three Löb conditions are precisely:

- L1: $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ (**BoxMP**);
- L2: $\Box A \rightarrow \Box \Box A$ (**Four**);
- L3: global proofs lift under $\Box$ (**Nec**).

The modal rule of the concrete calculus supplies these conditions. Its fixed-point constructor supplies $\text{fp } x.A \sim A(\text{fp } x.A)$ for every formula modalized in x ; Proposition 4.3 even gives infinitely many inequivalent Gödelian and Henkinian fixed points.

3.3 The precise missing morphism

Contraction would be a family

$\text{delta}_A : A \rightarrow A \text{ tensor } A$.

Restricted $\Box$ -contraction is only the family

$\text{delta}^\Box_A : \Box A \rightarrow \Box A \text{ tensor } \Box A$.

Beklemishev-Shamkanov Remark 3.9 shows that this restricted family, together with correspondingly weakened APS assumptions, is already enough for their G2 theorem. Their concrete system deliberately has no such family. In sequent form, one cannot uniformly pass from

$\text{Gamma}, \Box A, \Box A \vdash B$

to

$\Gamma, \Box A \vdash B$.

This is why the concrete S can satisfy L1-L3 and have fixed points while Proposition 4.1 proves that the formalized-G2 arrow is absent:

$\Box(\Box \text{bottom} \rightarrow \text{bottom}) \not\rightarrow \Box \text{bottom}$.

This also explains why an affine surface syntax is not enough: B-S note that an affine Peano arithmetic can still admit $\Box$ -contraction. The categorical test is the existence of $\delta^{\Box}$, not the absence of an unrestricted contraction rule.

3.4 Why this is not self-justification

Their Section 6 says explicitly that the example is not a counterexample to non-formalized G2, because

$\Rightarrow \text{not } \Box \text{bottom}$

is not provable. Categorically, $K_S(I, \text{Con})$ is empty. The example therefore occupies the cell

fixed points + HBL + no BoxContr + no Con point.

It proves that the HBL triple is insufficient once the ambient structural category ceases to be cartesian. It does **not** provide a system that asserts its consistency and remains consistent.

4. Willard: cartesian logic without internal uniform proof composition

4.1 Two levels that must be kept separate

Willard's generalized arithmetic is a pair (α, D) of an axiom basis and a deductive apparatus. Its categorical presentation must distinguish:

1. the external category in which mathematicians can manipulate finite proof codes; and
2. the internal syntactic category of operations and entailments whose totality the weak theory α itself can prove.

Let C_α be the cartesian syntactic category of α : objects are definable contexts and arrows are equivalence classes of functional relations whose totality and functionality α proves. Its predicate doctrine

$P_\alpha : C_\alpha^{\text{op}} \rightarrow \text{Pos}$

sends a context to its preorder of formulas modulo provable equivalence; reindexing is substitution. Finite meets and existential quantification give the usual first-order hyperdoctrine structure. The proof relation is a predicate

$\text{Prf}_{\{\alpha, D\}} \text{ in } P_\alpha(\text{Sent } x \text{ N})$.

For a fixed sentence Φ , pull it back along its code to obtain the proof predicate P_Φ over N , and put

$\Box_D \Phi := \text{exists } p. \text{Prf}_{\{\alpha, D\}}(\text{code}(\Phi), p) \text{ in } P_\alpha(1)$.

The terminal fiber $P_\alpha(1)$ is a genuine thin category: an arrow is an implication proved by α , and proved implications compose. This repairs the earlier draft's invalid "wide class of affordable arrows", which was expressly not closed under composition and hence was not a category.

The ordinary classical first-order logic of $IS(A)$ supplies cartesian context management, so contraction and the diagonals δ_X are present. Willard does not use the B-S escape hatch.

4.2 HBL (2) as uniform evaluation

For fixed sentences Φ, Ψ , HBL (2) has the following proof-witness form in the fiber over $N \times N$:

$\text{Prf}(\text{code}(\Phi), x) \text{ and } \text{Prf}(\text{code}(\Phi \rightarrow \Psi), y) \\ \rightarrow \text{exists } z. \text{Prf}(\text{code}(\Psi), z)$.

Its universal closure is equivalent, by ordinary first-order quantifier rules, to the terminal-fiber **BoxMP** entailment

$\Box \Phi \text{ and } \Box(\Phi \rightarrow \Psi) \rightarrow \Box \Psi$.

A stronger but familiar witness for this entailment would be a proof-combiner map

`comp_Phi,Psi : P_Phi x P_(Phi -> Psi) -> P_Psi.`

Van Dijk-Oldenziel Proposition 6.8 obtains exactly the analogous uniform operation from evaluation on internal Hom objects.

Willard’s 1993 technical report, printed p. 12, identifies the breach for Type-A **IS(A)**: the system can prove the required composition statement only when the proof-code variables **x** and **y** are fixed constants. In categorical terms, pulling the witness entailment back along a named point

`<x-bar,y-bar> : 1 -> N x N`

can yield a proved instance, while there is no entailment uniformly over the whole product. “Every named instance” does not assemble into the uniform evaluation morphism. This is analogous to, but not an identification with, the local-versus-global existence warning in van Dijk-Oldenziel Remark 5.12.

Accordingly, for the Type-A tableaux line the documented failure is the uniform witness entailment, and hence

`BoxMP_Phi,Psi` (HBL condition (2)),

not contraction. This document does not claim that every Willard configuration fails the same condition. The source corpus identifies condition (2) for **IS(A)**; for the separate Hilbert/no-growth line, the exact member of the HBL triple is not pinned down in the audited sources. External consistency plus a self-consistency theorem forces some premise of the applicable G2 package to fail, but it does not identify which one.

4.3 Selected diagonalization without total substitution

Willard’s fixed point exhibits the same pointwise/uniform split. Let

`Subst_i -> N x N`

be the definable relation saying that the second code results from substituting the first code into itself. The technical report, printed p. 37, states that **IS(A)** is too weak to prove

`forall x. exists y. Subst_i(x,y),`

but for every fixed numeral **k-bar** it can prove

`exists y. Subst_i(k-bar,y).`

Categorically, the projection `Subst_i -> N` is not proved internally to be a cover and has no proved global choice map, while the particular named fiber needed for Group-3 is inhabited. This yields the selected fixed point

`SelfRef(alpha,D) <-> not exists p. Prf_{alpha + SelfRef(alpha,D),D}(code(bottom),p).`

This is enough for the self-referential axiom. It is not the universal fixed-point property of van Dijk-Oldenziel Lemma 6.12, and no such universal claim is made here.

4.4 The positive data that make an SJAS

Let

`alpha* := B + SelfRef(B,D).`

Because `SelfRef(B,D)` is an axiom of **alpha***, its consistency proposition has a global proof in the syntactic doctrine:

`self : 1 -> Con(alpha*,D).`

That alone would be cheap: an inconsistent theory has such a point too. Willard’s Proposition 1 supplies the other half—under its stated hypotheses on the base and tableaux apparatus, **alpha*** is externally consistent. The Codification records its proof status as **full**.

For Type-A **IS(A)** the resulting categorical profile is therefore

`selected FP + cartesian contraction + Con point + external consistency + no uniform BoxMP.`

The HBL/Löb composite cannot turn **self** into a contradiction because its evaluation leg is not an internal morphism. This is stronger than B-S’s negative result: Willard supplies the desired consistency point and proves that it lives in a nondegenerate theory.

For later Willard systems using `SelfCons_0` or `SelfCons_1`, replace `Con` by the precisely coded Level-0 or Level-1 consistency predicate. One must not move between those predicates without rechecking the corresponding proof relation and consistency theorem.

5. The comparison square

The systems differ on two independent axes.

	Structural diagonal on boxed hypotheses	Uniform internal HBL evaluation
AU	present automatically ($x \rightarrow \langle x, x \rangle$)	present by Proposition 6.8
B-S s	absent ($\Box A \not\rightarrow \Box A \text{ tensor } \Box A$)	present as L1
Willard Type-A IS(A)	present in ordinary classical logic	absent internally ; fixed proof-code instances only

The fixed-point axis is separate again:

	Strength of diagonalization
AU	every endomorphism $T : P1' \rightarrow P1'$ (Lemma 6.12)
B-S s	every syntactically modalized fixed-point expression
Willard IS(A)	the selected Group-3 instance; total substitution is not internally proved

Finally, self-justification adds positive and semantic data:

	$1 \rightarrow \text{Con}$	external consistency
AU with full package	ruled out when consistent	hypothesis of G2
B-S s	absent by their §6	not the advertised result
Willard IS(A)	present as Group-3	proved under Proposition 1's hypotheses (full)

These tables should not be collapsed into a single “weakness” ordering. B-S weaken structural logic while keeping the modal derivability rules. Willard keeps structural logic and weakens what arithmetic can certify uniformly about its proof predicate. The systems meet only at the consequence: one leg of the G2 composite is missing.

6. What has and has not been proved

Proved in this comparison. From the source definitions, the three presentations instantiate the common interface as tabulated; the missing morphism in each case is identified; and Theorem 1.1 explains why each missing leg blocks the applicable HBL-G2 derivation. The B-S and Willard failure points are not identified with one another.

Cited rather than re-proved. B-S cut admissibility, failure of formalized G2, and non-provability of `not  $\Box$ bottom`; Willard’s consistency-preservation theorems and the exact Type-A HBL (2) failure; and the AU construction and Löb proof. Their proof status and image anchors are recorded in [VERIFICATION.md](#).

Not claimed.

- `IS(A)` is not shown to be an arithmetic universe or a sub-AU of `U_0`.
- The B-S monoidal category is not cartesian, and completing it cartesianly would add the very contraction whose absence is the theorem.
- The assignment $A \mapsto \Box A$ need not be an endofunctor in Willard’s internal doctrine; failure to transport arrows uniformly is the point.

- A failure of HBL or contraction is necessary to block this G2 proof but is never sufficient for self-justification.
 - The exact HBL breach is not asserted uniformly across every Willard deduction method.
-

7. Primary-source anchors

- J. van Dijk and A. Gietelink Oldenziel, *Gödel incompleteness through Arithmetic Universes after A. Joyal*, arXiv:2004.10482: Lemma 5.15; Theorem 5.20; §6 Definitions 6.1-6.3, Propositions 6.5-6.10, Lemma 6.12, Theorem 6.13, Remark 6.14. Local witness: [AU source PDF](#).
- L. Beklemishev and D. Shamkanov, *Some abstract versions of Gödel's second incompleteness theorem based on non-classical logics*, arXiv:1602.05728: Definitions 2.1, 2.3, 3.1-3.4, 3.7; Proposition 3.8; Remark 3.9; Propositions 4.1 and 4.3; Theorem 5; §6. Local witness: [B-S source PDF](#).
- D. E. Willard, 1993 technical report: printed p. 12 (fixed-witness-only HBL composition), printed p. 37 (fixed-instance substitution), Appendix A (Group-3). Local witness: [1993 technical report PDF](#).
- D. E. Willard, *Self-verifying axiom systems, the incompleteness theorem and related reflection principles*, JSL 66 (2001): Theorem A.1 and justification, pp. 44-45; consistency results as indexed in the Codification. Local witness: [2001 paper PDF](#).
- Corpus extraction and proof-status records: [1993 extraction](#), [2001 extraction](#), and the [result registry](#).